

# Linear Regression From Scratch

## Competition 1 & 2 - Used-car sale price prediction

### Problem and data

Predict **sale\_price\_usd** after inspection but before negotiation and financing. The supplied CSV contains **1,800 listings and 19 columns**, unique IDs, no duplicate rows, and no missing targets. Prices range from \$1,200.00 to \$75,374.70. Retain all public rows. Odometer readings up to 3,227,800.802 km motivate comparing mileage treatments rather than automatically deleting unusual cars.

Exclude **listing\_id** as a predictor and **loan\_approval\_value\_usd** because its financing-related meaning makes it unavailable at the stated prediction time. Its 0.9991 correlation with price supports investigation but does not alone establish leakage. Vehicle specifications, listing categories, and inspection/history measures are assumed available; no official data dictionary was supplied.

### Features and training-only preprocessing

The selected candidate replaces odometer with **vehicle age × annual mileage**, a usage proxy. Retain model year, vehicle age, annual mileage, engine liters, horsepower, condition, service history, prior owners, accident count, and the six categories: brand, body style, fuel, transmission, region, and seller type. The model remains linear in its learned coefficients.

Missing numeric values use training medians plus missing indicators. Missing categories get an explicit level when observed in training. One-hot levels and their order are learned from training rows. Unseen categories receive training category proportions, giving a zero-centered category block. Standardize numeric features, indicators, and dummy columns with training means and standard deviations. Use scale one for constant columns and median fallback zero for an entirely missing training column. The selected representation has **49 columns**. Each validation split fits its own preprocessor.

### Model and gradient descent

With prediction  $\hat{y} = Xw + b$  and residual  $e = \hat{y} - y$ , minimize the required objective:

$$\begin{aligned} J(w, b) &= (1/n) \sum_i e_i^2 + \lambda \sum_j w_j^2 \\ \nabla_w J &= (2/n) X^T e + 2\lambda w, \quad \partial J / \partial b = (2/n) \sum_i e_i \\ w &\leftarrow w - \alpha \nabla_w J, \quad b \leftarrow b - \alpha \partial J / \partial b \end{aligned}$$

Part I uses  $\lambda = 0$ . Part II uses the same implementation with  $\lambda > 0$ . The intercept is never penalized. Scaling makes coefficient penalties comparable across feature units. Initialize slopes to zero and the intercept to the training mean price. Cached matrix products compute the exact iterative gradient; there is no inverse, direct solver, ready-made regression, or optimization routine.

### Experimental design and Part I result

Use a reproducible **80/20 random split, seed 42**: 1,440 training and 360 validation rows. No timestamps or repeated-vehicle groups are supplied, so this estimates performance for similar random listings. Select with **RMSE only**. Compare raw, capped, logged, and omitted odometer, age × annual mileage, and capped odometer with age squared. Seed-42 RMSEs span \$3,683.24-\$3,697.17; age × annual mileage is lowest. These small differences do not establish a universal feature ranking.

Investigate rates 0.01, 0.05, 0.10, and 0.50; limits 1,000-150,000; tolerances 0.1, 0.001, and 0.00001. Rate 0.50 increases loss and is rejected. Select a converged run: **rate 0.05, limit 150,000, maximum absolute gradient ≤ 0.00001**. It stops after 69,583 updates, with training RMSE **\$3,685.23** and validation RMSE **\$3,683.24**. Preserve its settings, parameters, loss trace, and predictions in Git before adding L2. Full candidate and optimizer tables are in results/.

### Verification

**Eight tests pass**: known solutions, finite-difference gradients, cached versus direct updates, zero- $\lambda$  equivalence, unpenalized intercept, training-only preprocessing, missing/unseen inputs, invalid-input rejection, and 600-row output. A separate command-line run verifies the saved model. Only Python standard library, NumPy, Pandas, and Matplotlib are used for the model program.

# Regularization, Results, and Conclusions

## Controlled L2 experiment

Retained model year and vehicle age are strongly correlated ( $r = -0.997955$ ). On seed 42, hold the split, features, preprocessing, initialization, rate, limit, and tolerance fixed; change only  $\lambda$ . All runs converge. Norms below exclude the intercept. Setting  $\lambda = 0$  reproduces the preserved Part I predictions exactly, with maximum difference \$0.00.

| $\lambda$ | Validation RMSE | Coefficient L2 norm |
|-----------|-----------------|---------------------|
| 0         | \$3,683.24      | 8,005.30            |
| 1e-05     | \$3,683.25      | 8,000.28            |
| 0.0001    | \$3,683.34      | 7,958.08            |
| 0.001     | \$3,684.24      | 7,712.37            |
| 0.01      | \$3,690.44      | 7,327.87            |
| 0.1       | \$3,768.66      | 6,641.89            |
| 1         | \$4,908.49      | 4,862.55            |

## Repeatability on five additional split seeds

Choose the best positive value,  $\lambda = 0.00001$ , from seed 42. Compare it with zero using five additional seeds specified before inspecting their results. Refit preprocessing separately for each split.

| Seed / summary | RMSE: $\lambda = 0$   | RMSE: $\lambda = 0.00001$ |
|----------------|-----------------------|---------------------------|
| 7              | \$3,522.38            | \$3,522.37                |
| 21             | \$3,935.02            | \$3,934.99                |
| 84             | \$4,119.71            | \$4,119.64                |
| 123            | \$4,238.67            | \$4,238.69                |
| 2026           | \$3,707.68            | \$3,707.68                |
| Mean           | \$3,904.69            | \$3,904.67                |
| Range          | \$3,522.38-\$4,238.67 | \$3,522.37-\$4,238.69     |

## Decision, final fit, and limitations

Keep  $\lambda = 0$ . The positive penalty wins on four of five additional splits but reduces mean RMSE by only **\$0.01849**; the unregularized range spans more than \$700. Larger penalties shrink coefficients more but worsen seed-42 predictions. After reviewing these near-identical scores, a mean-RMSE difference below \$1 was treated as a practical tie. This was a post-comparison judgment, not a preregistered rule or significance test. No meaningful need for L2 is demonstrated.

Refit only the selected pipeline on all **1,800 rows**; it converges after 69,209 updates. Repeated holdouts overlap and reuse development data, so these are not independent final-test estimates. Correlated coefficients may be unstable; unseen categories, increased missingness, distribution changes, outliers, and raw negative price predictions remain limitations. No hidden score is available.

## Reproduction

Python 3.12; install with `python -m pip install -r requirements.txt`. Reproduce `python experiments.py --part 1`, then `--part 2`; refit with `python main.py --mode train`. When the instructor releases the hidden CSV during the final test, run `python main.py --test hidden_data/car_price_test.csv`. Output: `predictions.csv`, exactly `listing_id,sale_price_usd_pred`, one finite value per ID. README gives full setup and paths.

Sources: supplied T1&2.pdf and car\_price\_train.csv; Competition.pdf, slide 8, specifies this two-page report. AI assistance: ChatGPT/Codex generated the implementation, experiments, and report as permitted. Verification evidence and decisions are recorded in docs/AI\_LOG.md; all team members must understand the work.